Vibe Shuffle: Relative Valence–Arousal Adaptation for Music Selection

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Abstract

Music recommendation usually models durable preferences, although a suitable track changes with context. *Vibe Shuffle* is a browser application that adapts the next-song choice from camera and cardiac signals. Recent observations are expressed as changes from short personal reference periods. Facial cues supply relative Valence evidence; heart rate and beat-to-beat variability supply relative Arousal evidence. During playback, these estimates form a trajectory in Valence–Arousal space. The trajectory mean selects a recommendation region, and the policy chooses the nearest unplayed song in the corresponding Valence–Energy region of a fixed catalog. A local export records the target, region, song distance, derived signals, and ratings. Raw camera frames, Bluetooth packets, facial-reference values, and point-level trajectories remain unexported. An exploratory crossover left the effects on mood fit and liking unresolved. A larger study with complete trial-level records is needed to estimate the policy's effect.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, baseline-relative adaptation

1 Introduction

Music choice is contextual. A track that suits exercise may be unwelcome during a quiet break. People use music to regulate mood and activation [10, 24]. Most recommenders prioritize stable signals such as listening history, favorite artists, similar users, and item features. These signals characterize usual preference and capture momentary fit only indirectly, a known challenge in contextual music recommendation [11, 25].

Vibe Shuffle uses browser camera input and optional wearable data to adapt the next-song choice to recent within-listener change. Full affect recognition would require reliable links among facial behavior, physiology, context, and subjective experience. A camera captures visible movement. Cardiac timing also varies with respiration, posture, activity, medication, and fitness. Neither channel directly identifies emotion.

During setup, the listener supplies short personal references. Later facial and cardiac observations are expressed as changes from those references. The horizontal axis represents relative Valence evidence, and the vertical axis represents relative Arousal evidence. Each point falls into a Tense, Energetic, Melancholic, or Calm recommendation region. The names identify catalog regions. The first track precedes a complete trajectory; subsequent choices use the preceding track's summarized path.

The catalog uses corresponding Valence and Energy coordinates. Vibe Shuffle averages the recent listener trajectory, retains unplayed songs from the matching region, and chooses the nearest one. A Random condition draws from the same remaining catalog without using the trajectory. The adaptive rule is *relative signals* → *recent point* → *region* → *nearest available song*.

Vibe Shuffle implements a baseline-relative signal representation, an observation-to-trajectory mapping, a four-region nearest-song policy over 100 tracks, participant-facing condition masking, and a local row-level export. In the 15-participant crossover, the mean mood-fit difference was +0.56 points (95% CI $[-0.21, 1.33]$).

2 Background

2.1 A relative valence–arousal coordinate

The circumplex model represents affect along two continuous dimensions: *valence*, from unpleasant to pleasant, and *arousal*, from low to high activation [21, 23]. Thayer's account of energetic and tense arousal offers a complementary vocabulary for changes in activation and mood [27]. The interface uses the Valence–Arousal axes and selected category labels from these models. Both dimensional and discrete models appear in music-emotion research, and results depend on the selected model and annotations [6].

Figure 1 juxtaposes the two affect vocabularies with group-level HRV-frequency evidence used to motivate an exploratory diagnostic signal.

The interface maps sensor-derived coordinates onto the two-axis recommendation space. It does not output the emotion terms or coherence profiles shown in Figure 1.

Each listener point (V, A) records change from personal reference measurements. The center marks the reference. More positive facial display moves the point right, and greater cardiac activation moves it upward. Facial movements vary by person and context and do not uniquely identify emotions [4]. Body context also shifts facial judgments, and Valence–Arousal relations vary across people and cultures [1, 13].

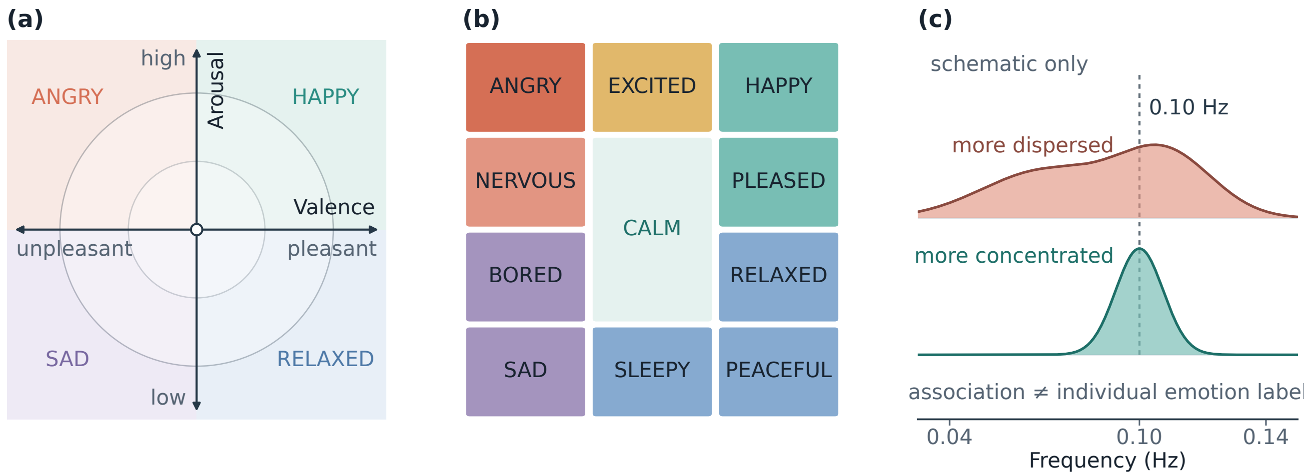

Figure 1: Conceptual foundations. Panels (a)–(b) are original redrawings adapted from Helmholz et al. [8], who operationalized Russell's circumplex [23] and Thayer's mood–arousal model [27]. Panel (c) is an original schematic adapted from the group-level frequency distinction reported by Balaji et al. [3]; it contains no empirical observations.

2.2 Connecting listener state to music

Songs can be placed in a parallel two-dimensional space. Song *Valence* describes conveyed positivity, whereas *Energy* describes perceived intensity and activity. Prior emotion-oriented music interfaces have used these attributes as musical counterparts to affective valence and arousal [8]. Vibe Shuffle compares listener Valence with song Valence and listener Arousal with song Energy. Musical Energy and physiological Arousal remain distinct constructs. Continuous music-emotion coordinates also depend on subjective annotations [29].

Splitting each axis at 0.5 yields four recommendation regions. The listener's relative position selects a region. Quadrant membership determines eligibility, and Euclidean distance ranks the eligible songs.

2.3 Heart-rate variability as relative evidence

Heart-rate variability (HRV) describes variation between successive heart beats. The root mean square of successive differences (RMSSD) is a widely used time-domain summary of short-term RR-interval variation [26]. Heart rate and RMSSD depend on respiration, movement, posture, fitness, medication, sensor quality, and other factors. These measures lack emotion-specific interpretation. Recommended practice includes standardized conditions, artifact review, matched durations, and respiration assessment [14].

Vibe Shuffle uses cardiac measurements as baseline-relative evidence for the Arousal axis. Heart rate above the reference and RMSSD below it contribute upward evidence; changes in the opposite direction lower it. The cardiac rule has not undergone clinical or psychological validation.

3 Related Work

3.1 Emotion-aware music selection

Music can evoke, express, maintain, or help regulate affect through several mechanisms [10]. Emotion-aware music systems use listener-provided or inferred context to constrain the next track.

Helmholz et al.'s *Feel the Moosic* uses a two-axis interface. Its color-coded representation simplifies the Russell and Thayer models. A listener selected a position, and the prototype requested songs within corresponding Spotify Valence and Energy ranges [8]. Vibe Shuffle retains the two-axis mapping and derives the listener coordinate from baseline-relative camera and cardiac evidence. It summarizes a trajectory over the listening window and selects an unplayed catalog track near the mean coordinate. The interface reports the resulting target, region, and distance.

3.2 Personal and multimodal affective recommenders

Prior systems have used personal physiological or camera cues to select music. Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player toward a goal [9]. Ayata et al. compared machine-learning models that predicted Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements before recommending music [2]. Tran et al. used a facial image to estimate a point in Valence–Arousal space and retrieved nearby songs in Spotify feature space [28].

Vibe Shuffle applies a fixed rule that maps recent within-person changes to a catalog coordinate. The study uses participant/session-level recommendation ratings, and the export stores the trajectory summary, target region, and distance but not point-level samples.

3.3 HRV and coherence evidence

HRV provides useful information about cardiac dynamics, but its interpretation depends on measurement duration and context [26]. Balaji et al. analyzed 1,884,216 mobile HRV-biofeedback sessions from 70,041 users and reported aggregate associations between coherence measures and emotional labels entered by users after their sessions [3]. Sessions labeled with positive emotions had higher mean coherence and more stable coherence frequencies than sessions labeled with negative emotions. The study was an observational analysis of 3–15 minute biofeedback practice, lacked pre-session emotion reports, and did not establish a per-window emotion classifier.

Vibe Shuffle exports a custom normalized spectral-peakedness proxy for later diagnostic analysis. This field differs from the coherence measure analyzed by Balaji et al. and has no role in coordinate, region, or song selection. Selection uses the baseline-relative heart-rate and RMSSD estimate. The spectral-peakedness proxy has not been validated as either a coherence measure or a basis for individual emotion inference.

3.4 Explanations and user experience

Recommender explanations can target transparency, trust, effectiveness, or decision support [20]. The Vibe interface displays the target, region, eligibility status, and distance. Perceived quality, effort, privacy, and context remain separate aspects of user experience [12].

3.5 Comparison with prior systems

Vibe Shuffle differs from the cited systems in two implementation choices: a fixed baseline-relative mapping and a Random condition drawn from the same frozen catalog. The export records the coordinate, region, eligibility, and distance used for each recommendation.

4 System Design and Implementation

4.1 System overview

Vibe Shuffle is a client-only React/Vite application. Camera input, an optional Bluetooth Heart Rate Service device, and Spotify playback are integrated within one browser session. The application has no study database or recommendation server. Adaptation occurs between tracks: the latest listening window is summarized before the next track is selected. This interval-level update produces one decision per track and prevents brief fluctuations from changing playback immediately.

Figure 2 summarizes the process. Camera and cardiac measurements are expressed relative to personal reference values. During confirmed playback, successive coordinates form a trajectory. Its mean selects a region and a nearby song. The participant rates the track after playback.

4.2 Relative Valence from camera cues

The browser runs MediaPipe Face Landmarker locally [16]. It returns facial blendshape activations such as smile, cheek, brow, eye, jaw, frown, and mouth movement. A fixed rule combines a selected subset into positive, negative, and tension-related scores. Vibe Shuffle defines these scores; they are absent from the MediaPipe output.

A 14-second calibration captures a short facial reference, and a slow running baseline continues to adapt. Later observations are debiased against this reference and smoothed across frames. Positive facial evidence moves Valence to the right; negative or tension-related evidence moves it to the left. If no face is detected, Valence returns to the neutral center. The current heuristic treats visible movement as additional positive-Valence and upward-Arousal evidence. Dancing, repositioning, lighting changes, and camera noise can produce the same signal. The movement rule lacks validation as a measure of engagement or affect.

4.3 Relative Arousal from cardiac timing

The optional wearable provides device-produced beats per minute and, when available, RR intervals—the elapsed time between successive beats. The application receives device-produced summaries and has no access to a raw electrocardiogram or the device's peak-detection process. Implausible and abruptly changing RR intervals are rejected before short-term variability is calculated.

The primary variability feature is the root mean square of successive differences (RMSSD). A 120-second reference stores the listener's median heart rate and RMSSD together with robust estimates of their variation. During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence. Heart rate receives the larger influence because RMSSD estimated from a short window is unstable; when the two disagree, the RMSSD contribution is strongly reduced. At least 20 accepted RR intervals are required for a usable full estimate. Muñoz et al. found high agreement between 120-second and longer RMSSD recordings under controlled resting conditions [18]. Their study did not examine short wearable windows during movement or Arousal estimation.

The implementation also exports SDNN and a custom normalized frequency-domain spectral-peakedness score, recorded in the CSV as the physiology coherence field. This experimental diagnostic differs from the coherence measure analyzed by Balaji et al. and has no role in region selection. Balaji et al. report group-level associations for a different coherence measure [3]. Vibe Shuffle's proxy has not been validated against that measure or against individual emotion labels. Song selection uses only the baseline-relative heart-rate and RMSSD estimate.

4.4 Trajectory aggregation

Whenever the fused live coordinate changes during confirmed playback, Vibe Shuffle appends a point to the current trial buffer. Facial cues primarily control Valence. Usable cardiac evidence defines the Arousal base; visible movement can increase it. Without usable cardiac evidence, camera movement provides an upward-only fallback. If both channels are unavailable, the coordinate returns to $(0.5, 0.5)$.

The trial trajectory is the sequence

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the application uses the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m} \sum_{i=1}^{m} (V_i, A_i)$$

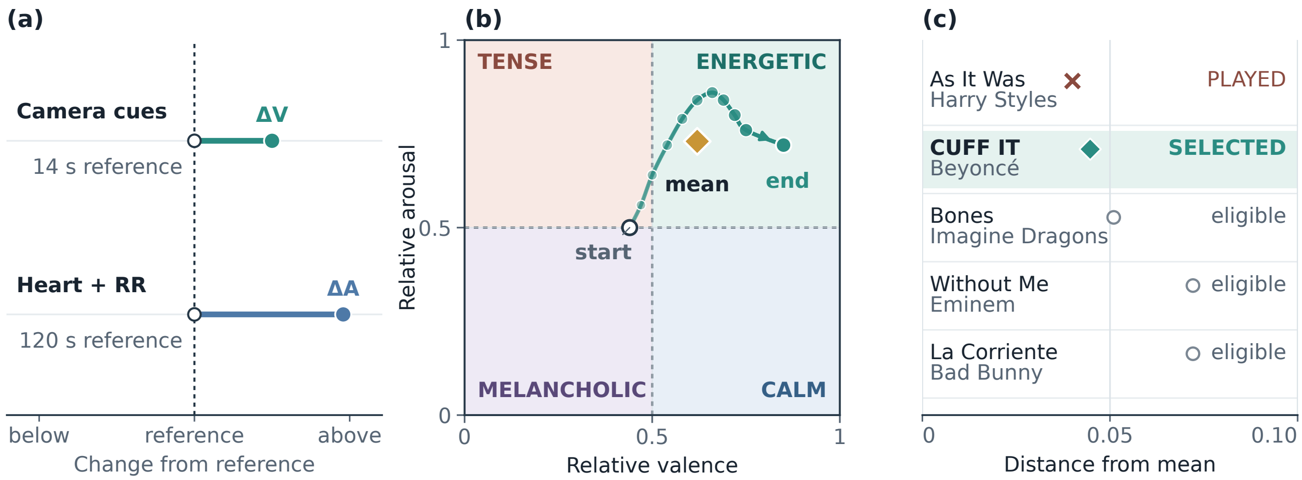

Figure 2: Relative sensing, trajectory, and selection in Vibe Shuffle. (a) Camera and cardiac inputs are expressed as signed changes from personal references. (b) A curved Valence–Arousal trajectory is summarized by its arithmetic mean (gold diamond); points grow and darken from the open start marker to the filled endpoint. (c) Within the matching region, *As It Was* is nearest but excluded as played (cross), so *CUFF IT* becomes the selected eligible track (filled diamond); open circles are alternatives. Panels (a) and (b) are schematics. Panel (c) uses catalog candidates, distances, and the repository's selection result. The trajectory illustrates the implemented aggregation rule. The logged coherence field is excluded from selection.

as the target for the next adaptive choice. Points are appended on coordinate changes, without a fixed sampling rate. The arithmetic mean is event-weighted. The export includes sample count and listening duration for review of this approximation.

4.5 Four regions and a fixed music space

Splitting both axes at 0.5 produces four recommendation regions: Energetic (higher Valence and Arousal), Calm (higher Valence and lower Arousal), Tense (lower Valence and higher Arousal), and Melancholic (lower on both axes). The frozen catalog contains 100 Spotify tracks, with 25 tracks in each corresponding Valence–Energy region. Track coordinates originate from historical Spotify features in a public dataset [17]. Energy serves as the catalog-side proxy for Arousal. The audio feature and physiological construct are not equivalent.

Both Random and Vibe exclude the current and all previously played tracks. Random ranks the remaining tracks with a session-seeded pseudorandom score. Vibe first retains available tracks in the target region and then selects the smallest Euclidean distance:

$$s^* = \arg\min_{s \in Q(\bar{V},\bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

The seed resolves only an exact distance tie. If a region has no remaining track, the policy falls back to the complete unplayed pool. The export records the target coordinate, signal source, distance, and region match. The random seed, personal facial-reference values, and individual trajectory points are absent. The first track is chosen from the live setup estimate before the 14-second facial reference is finalized. Later choices use the preceding listening window.

4.6 Data flow and local export

Spotify authentication uses Authorization Code with PKCE, and the Web Playback SDK provides playback for an allowlisted Premium account. Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. The locally downloaded CSV contains derived signal summaries, song metadata, selection fields, ratings, quality flags, pseudonymous identifiers, and timestamps. It is downloaded automatically at protocol completion and can also be downloaded manually. The file contains no images, landmark sets, raw waveforms, or audio. Its trajectory summaries allow review of the final target, but individual sensing steps cannot be replayed.

Authentication, asset delivery, and hosting require network requests to Spotify, MediaPipe, and GitHub Pages. Participant numbers and timestamps in the export are pseudonymous and should be handled as research data.

5 Exploratory Crossover Study

5.1 Protocol

The study compares the perceived appropriateness of the relative adaptive policy with Random selection from the same catalog. Each participant completes five tracks in each condition. Two masked protocols reverse block order, and the interface uses odd/even participant numbers to suggest an assignment. In participant-facing mode, the condition name is replaced by a protocol identifier. The experimenter can access the view toggle, protocol controls, and condition order in exported data. Masking depended on study administration and was not enforced by the interface. The study was not double blind. Practice and carryover remain possible in the paired design; reversing block order helps control those effects [7].

Table 1: Exploratory condition summaries ($N = 15$ participants). Confidence intervals are two-sided. The reported study tests used one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]

Each track plays for up to 60 seconds. Participants may select *Rate now*; the actual duration and an early-rating flag are preserved. After each track, the interface collects a seven-point liking rating, a seven-point mood-fit rating, and a categorical current-mood report. We designated mood fit as the primary exploratory outcome because it measures the proposed match directly. Liking was secondary and separates contextual fit from general song preference.

5.2 Available evidence

The supplied analysis report contains one five-trial condition mean per participant/session for 15 participants, together with paired tests and condition summaries. The 15 rows are frozen in the repository under generic analysis IDs. The protocol implies 75 planned ratings for each condition and outcome. Without the trial rows, completeness, exclusions, and within-session variation cannot be checked. Recruitment records, demographics, and the complete study-administration file are also unavailable. The available files limit the analysis to the 15 participant/session means. The human-study archive remains incomplete.

Paired differences are defined as Vibe minus Random. The analysis includes the published participant/session means, condition means and standard deviations, paired mean differences with two-sided 95% confidence intervals, and the originally supplied paired tests. No dated preregistration was available. We report the paired estimates, 95% confidence intervals, and supplied p-values as exploratory results [5, 19]; no confirmatory significance threshold was specified.

6 Results and Limitations

6.1 Mood-fit estimate and uncertainty

Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired mean difference was +0.56 rating points with a 95% confidence interval from −0.21 to 1.33. The supplied directional paired test yielded $t(14) = 1.56$, $p = .070$; the signed-rank sensitivity check yielded $W^+ = 74.0$, $p = .088$. The standardized paired effect was $d_z = 0.40$. Eight participant/session means were higher under Vibe, one was tied, and six were lower. The 95% confidence interval includes effects on either side of zero. Figure 3 plots the paired participant means and Vibe–Random differences.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09 with a 95% confidence interval from −0.52 to 0.71, $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. The liking confidence interval includes zero; the study did not specify an equivalence margin.

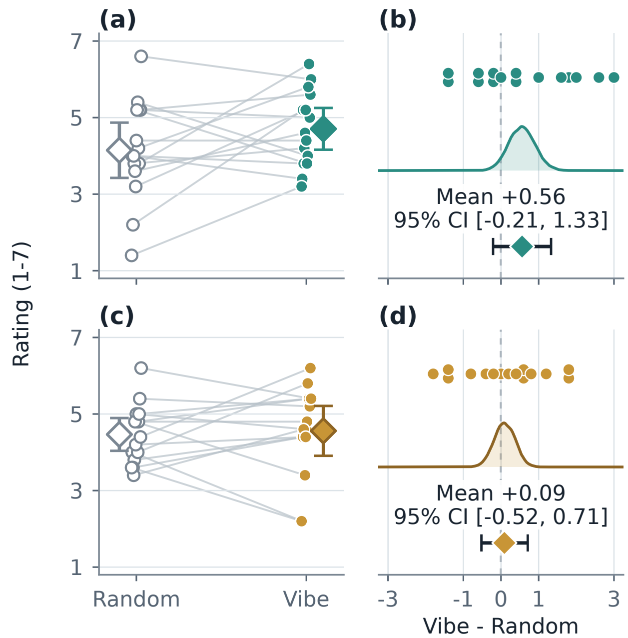

Figure 3: Participant-level paired outcomes. Panels (a)–(b) show mood fit; panels (c)–(d) show liking. Left panels connect Random and Vibe ratings and show condition means with 95% confidence intervals. Right panels show participant differences, bootstrap mean distributions, and paired mean 95% confidence intervals; both intervals cross zero.

6.2 Software checks

Core selection and signal-processing rules are implemented as pure JavaScript modules and can be tested without a camera, wearable, or Spotify account. The repository tests catalog size and region balance, CSV finiteness and quality flags, facial-rule behavior, RR filtering and baseline-relative cardiac behavior, and Random/Vibe selection. All 63 deterministic tests pass in the verified repository state. The suite tests software contracts; it contains no labeled affect data or independent sensor ground truth.

Both policies used the same playback path, rating questions, exposure count, and catalog. Their selected tracks could differ in artist, genre, language, familiarity, and popularity.

6.3 Measurement and study limits

The facial and cardiac mappings have not been validated against ground truth. Facial behavior cannot be reduced reliably to a universal readout of felt emotion [4]. Vibe's rule weights were not learned or validated on a labeled affect dataset. Lighting, pose, occlusion, morphology, culture, and deliberate display can all change the output.

Heart rate and RMSSD also vary with respiration, posture, speaking, movement, fitness, medication, contact quality, and vendor processing [22, 26]. The listening window is short for HRV, respiration is not measured, and the application receives processed

RR intervals without access to raw peaks. Personal baselining adjusts between-person levels but leaves within-session physiological changes and sensor artifacts in the estimate.

Movement can affect the camera channel, raise heart rate, and introduce RR artifacts at the same time. One behavior can be counted by both channels. The trajectory mean weights coordinate-change events instead of elapsed time, and the first selection may occur before both references are fully established. Because sensing is optional, participants may experience different evidence paths.

At the exact no-signal center, current code paths disagree on which region owns the boundary: the fallback label is Calm, whereas the numerical greater-than-or-equal rule assigns $(0.5, 0.5)$ to Energetic. This edge case should be unified before a confirmatory study.

The repository lacks raw trial rows, participant characteristics, recruitment and compensation details, the exact sensor model, room conditions, consent documentation, and the ethics procedure. No inference can be made about whether the undocumented procedures occurred. Without these records, recruitment, ethics, sensor setup, and trial-level data processing cannot be independently reconstructed.

6.4 Next validation steps

Validation should begin with controlled changes in expression, lighting, posture, movement, and respiration applied to the frozen facial and cardiac mappings. Release should then require successful calibration, timing, missing-data flags, and a complete ten-trial export. Before collecting a larger crossover sample, preregister the meaningful mood-fit effect, sample size, exclusions, primary paired analysis, order analysis, and sensor-quality checks. Base sample size on interval precision or a prespecified smallest relevant effect [15]; the pilot estimate is too uncertain to serve as the target.

Future protocols should specify whether the policy is intended to maintain, intensify, or shift the recent region. The outcome measure should match that stated goal.

7 Conclusion

Vibe Shuffle uses baseline-relative camera and cardiac observations to select the next song. It tracks recent movement in Valence–Arousal space, averages the trajectory, maps that point to one of four regions, and chooses the nearest eligible song. The exported target, region, and distance identify the rule used for each recommendation.

In the 15-participant crossover, average mood fit was 0.56 points higher under Vibe (95% CI $[-0.21, 1.33]$), and liking was 0.09 points higher (95% CI $[-0.52, 0.71]$). The current-mood reports were subjective ratings, and the software produced no objective emotion label. The stored target, signal-source summary, region, and song distance document the final decision; the export omits point-level trajectory samples.

A follow-up study should validate each sensor rule under controlled conditions, use time-weighted fusion, retain complete trial-level provenance, and preregister a larger paired comparison. The current system uses physiology only to shift the recommendation coordinate relative to a personal baseline.

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